

Harnessing Home IoT for Self-tracking Emotional Wellbeing: Behavioral Patterns, Self-reflection, and Privacy Concerns

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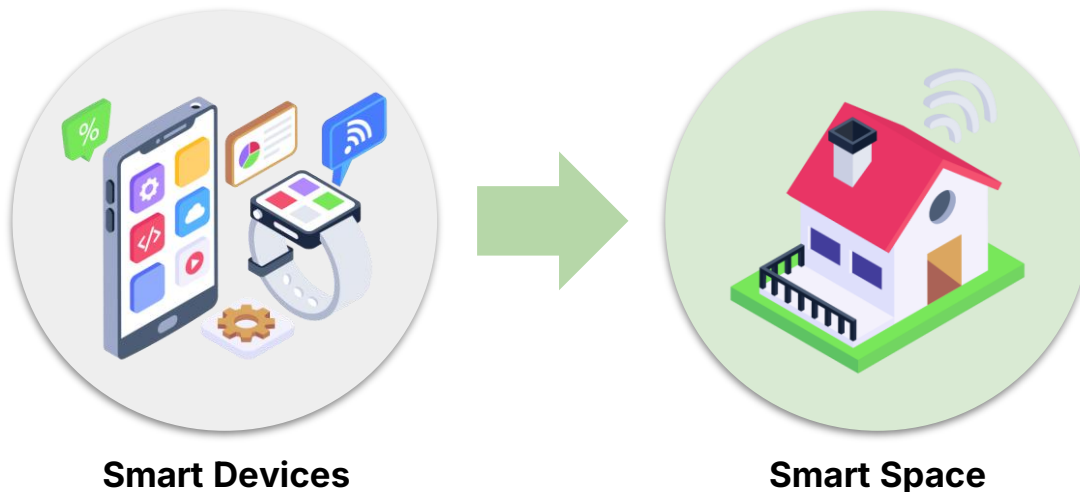
Interactive Computing Lab, KAIST



Self-tracking for Emotional Wellbeing

HCI/UbiComp research has explored **self-tracking for emotional wellbeing**, primarily through **mobile and wearable devices**

Recent advances toward Ambient Intelligence^[1] extends this vision, shifting self-tracking from **personal devices** to **environmental sensing**



[1] Mark Weiser. "The computer for the 21st century." ACM SIGMOBILE mobile computing and communications review 1999

Home as a Key Context for Emotional Wellbeing



Home Stay ↑

Longer time at home with poor mental health^[1]



Everyday Routines at Home

Daily routines at home strongly linked to mental health^[2]



Private & Natural Space

Captures natural behaviors in a comfortable environment^[3]

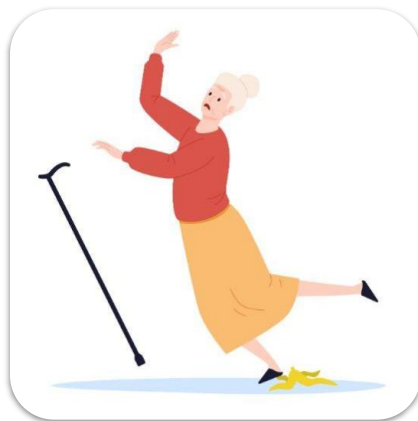
[1] Sheila J Linz and Bonnie A Sturm. "The phenomenon of social isolation in the severely mentally ill." Perspectives in psychiatric care 2013

[2] Sarris et al. "Lifestyle medicine for depression." BMC psychiatry 2014

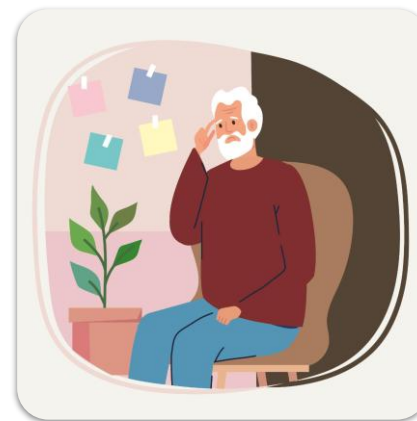
[3] Rooksby et al. "Personal tracking as lived informatics." CHI 2014.

Existing Research on Home IoT Sensing

Existing studies monitors **Activities of Daily Living (ADLs)** in home environments, focusing on **older adults' cognitive and functional health**^[1-4], with little attention to emotional wellbeing



Functional Health
(e.g., fall detection)



Cognitive Health
(e.g., early detection of dementia)

[1] Miao Yu et al. "A posture recognition-based fall detection system for monitoring an elderly person in a smart home environment." IEEE transactions on information technology in biomedicine 2012

[2] Daniele Riboni et al. "SmartFABER: Recognizing fine-grained abnormal behaviors for early detection of mild cognitive impairment." Artificial intelligence in medicine 2016

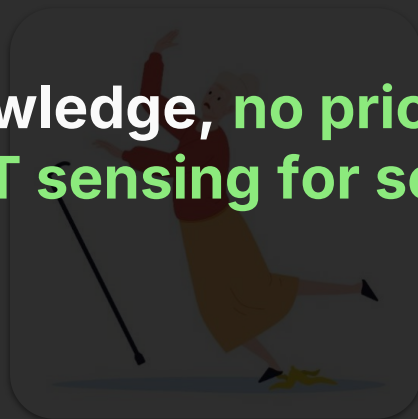
[3] Andreadis et al. "Dem@ Home: Ambient intelligence for clinical support of people living with dementia." European Semantic Web Conference 2016

[4] Ane Alberdi et al. "Smart home-based prediction of multidomain symptoms related to Alzheimer's disease." IEEE journal of biomedical and health informatics 2018

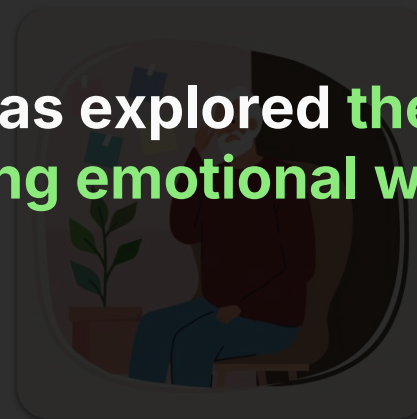
Existing Research on Home IoT Sensing

Existing studies monitor **Activities of Daily Living (ADLs)** in home environments, focusing on **older adults' cognitive and functional health**^[1-4], with little attention to emotional wellbeing

To our knowledge, no prior study has explored the potential of home IoT sensing for self-tracking emotional wellbeing



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(e.g., fall detection)



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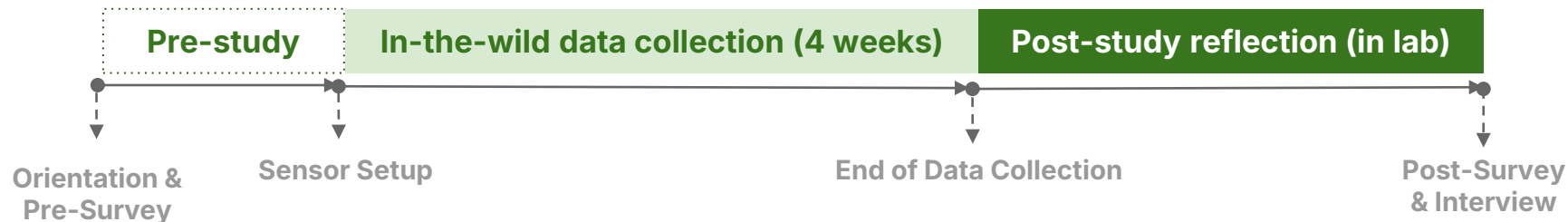
Research Goal

Explore how **home IoT sensing** can be used for **self-tracking emotional wellbeing**, focusing on **behavioral patterns, user perceptions, and privacy concerns**

Research Questions (RQs)

- 💡 RQ1. How is **home IoT data** linked to **everyday emotional wellbeing**?
- 💡 RQ2. How do users **perceive behavioral patterns** derived from **home IoT data**?
- 💡 RQ3. What are users' **privacy concerns** about using **home IoT data** for wellbeing tracking?

Overview of Study Phases



Pre-study

- Baseline mental health assessment survey (PHQ-9, GAD-7, PSS)
- Privacy acceptability survey (before data collection)

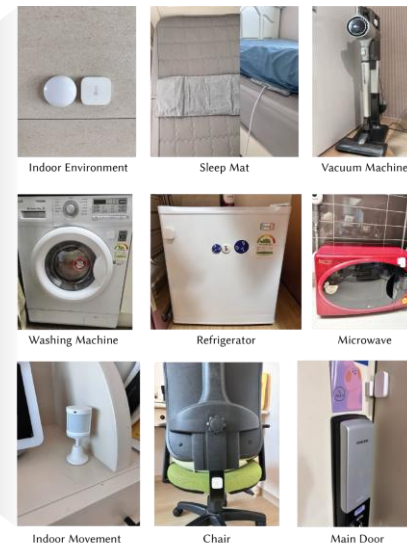
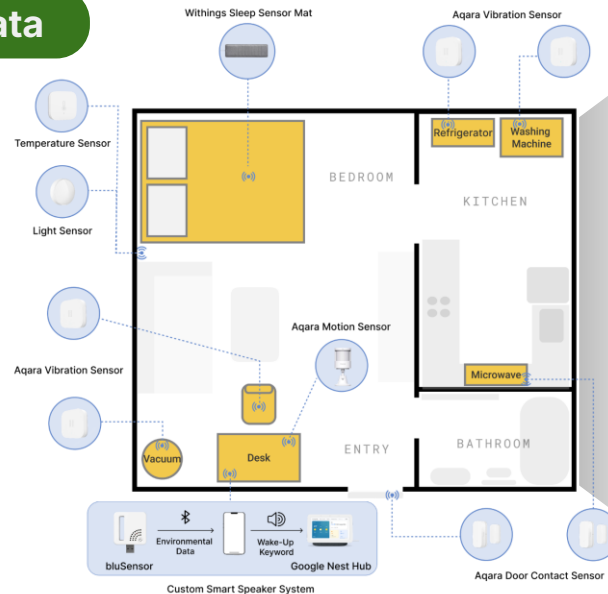
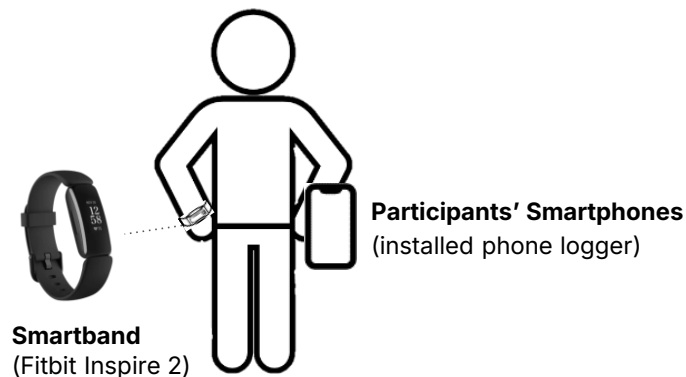
In-the-wild data collection (4 weeks)

- 20 single-person households (13 men, 7 women; avg age: 24.8), each living in a studio or one-bedroom unit
- Mobile, wearable, and home IoT sensor data, and self-reported emotional wellbeing survey via ESM

Post-study reflection (in lab)

- Baseline mental Self-reflection using visualization tool
- Privacy acceptability survey (after data collection)
- Exit interview

In-the-Wild Data Collection: Sensor Data



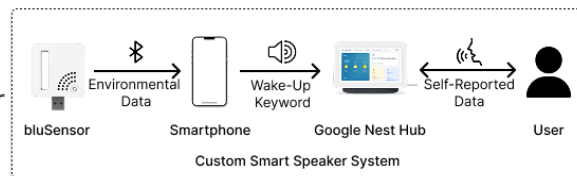
Mobile & Wearable

- Call & SMS metadata (frequency, duration)
- Step counts

Home IoT Sensors

- Sleep time
- Appliance usage
- Indoor movement
- Outgoing/arrival events
- Environment (brightness, temperature, humidity)

In-the-Wild Data Collection: Self-reported Data



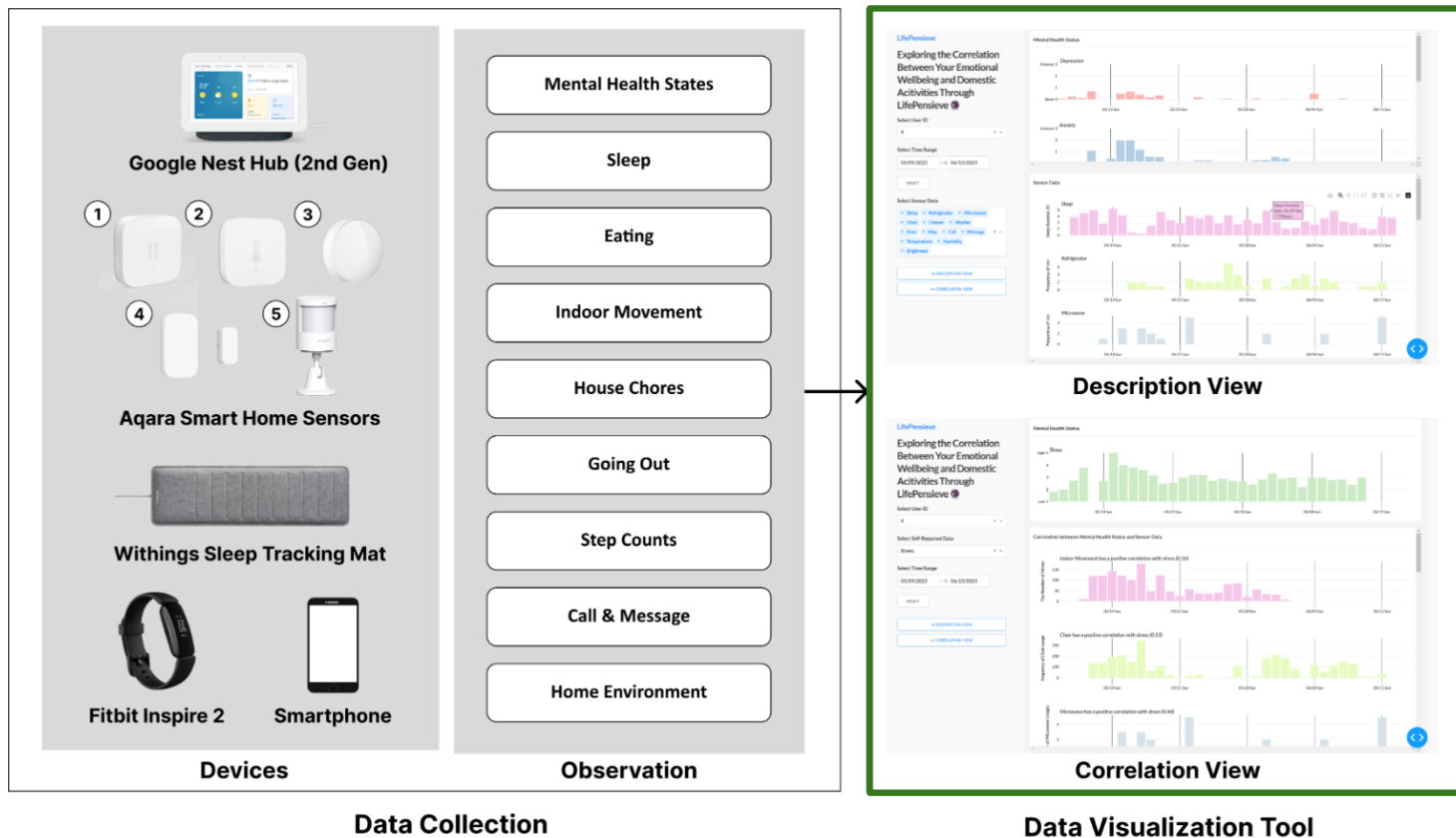
The image shows a smart speaker screen displaying a survey program. The text on the screen reads: "Survey Program", "Over the 1-2 hours before the survey, how often have you been bothered by little interest or pleasure in doing things?", and "Please tap the button or speak your answer". Below this, there are four circular buttons labeled 0, 1, 2, and 3, corresponding to the response options: "Not at all", "Slightly disrupted", "Frequently disrupted", and "Very frequently disrupted". At the bottom of the screen, there are navigation controls: a "Prev" button, a "1/7" indicator, and a "Next" button.

[Smart Speaker UI Example]

Mental Health Surveys Conducted via a Smart Speaker

- Every 90 minutes or at contextually opportune moments
 - Opportune moments: noise ↓, user nearby
 - Smartphone app activates speaker with a wake-up keyword
- Users answer **mental health questionnaires**
 - PHQ-2, GAD-2, Stress, Emotional Arousal & Valence
 - Response via **voice** or **touch**

Post-study reflection



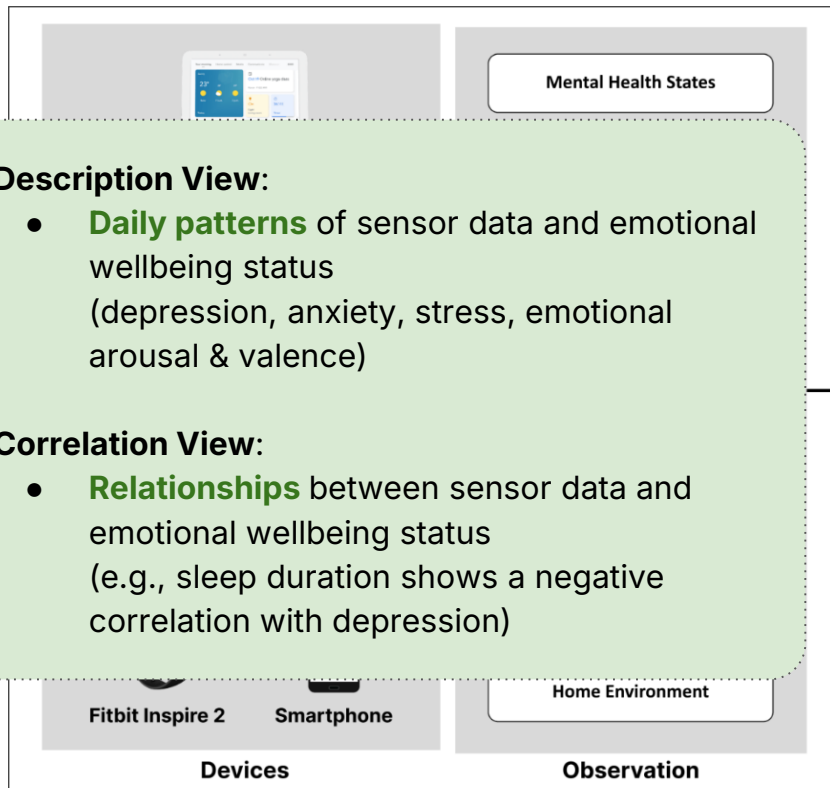
Post-study reflection

Description View:

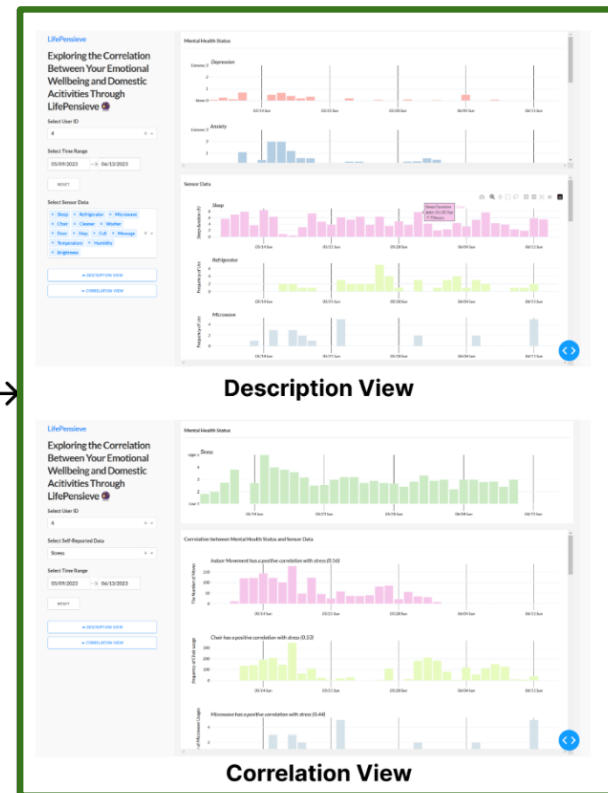
- **Daily patterns** of sensor data and emotional wellbeing status (depression, anxiety, stress, emotional arousal & valence)

Correlation View:

- **Relationships** between sensor data and emotional wellbeing status (e.g., sleep duration shows a negative correlation with depression)



Data Collection



Data Visualization Tool

Findings for RQ1. Exploring Home IoT Data Associated with Emotional Wellbeing

Multilevel Logistic Regression Analysis Results:

Reduced sleep duration and **higher indoor temperature** predicted **worse mental health**

Data Types	Depression	Anxiety	Stress	Emotional Valence	Emotional Arousal
Sleep duration	↓ ($\beta = -0.21^{***}$)	↓ ($\beta = -0.33^{***}$)	↓ ($\beta = -0.12^*$)	↑ ($\beta = +0.11^*$)	—
Temperature	↑ ($\beta = +0.77^{***}$)	↑ ($\beta = +0.74^{***}$)	—	—	—
Washer usage	↑ ($\beta = +0.22^*$)	—	—	—	↑ ($\beta = +0.17^{**}$)
Door open/close	—	—	↑ ($\beta = +0.10^{**}$)	—	—
Message counts	—	—	↑ ($\beta = +0.08^*$)	—	—
Other features*	—	—	—	—	—

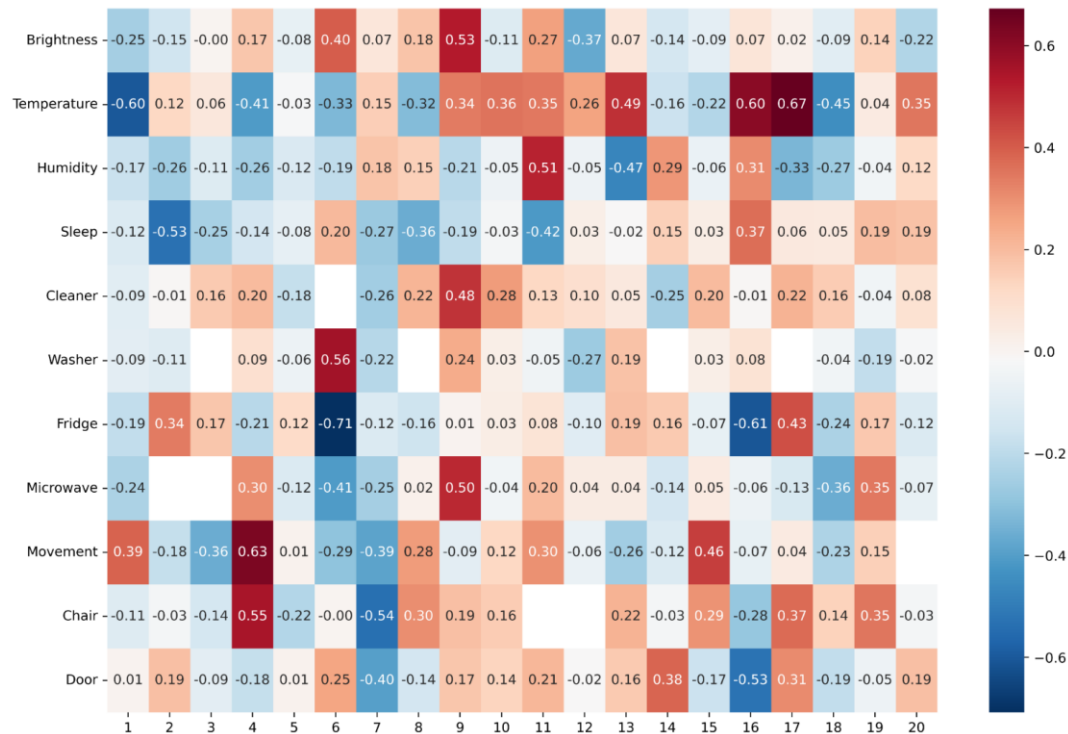
*Other features include call frequency, step count, brightness, humidity, and appliance or furniture usage (e.g., cleaner, fridge, microwave, chair)

Blue: Negative association (Higher value = Lower emotional score)

Red: Positive association (Higher value = Higher emotional score)

— : non-significant ($p \geq .05$)

Findings for RQ1. Exploring Home IoT Data Associated with Emotional Wellbeing



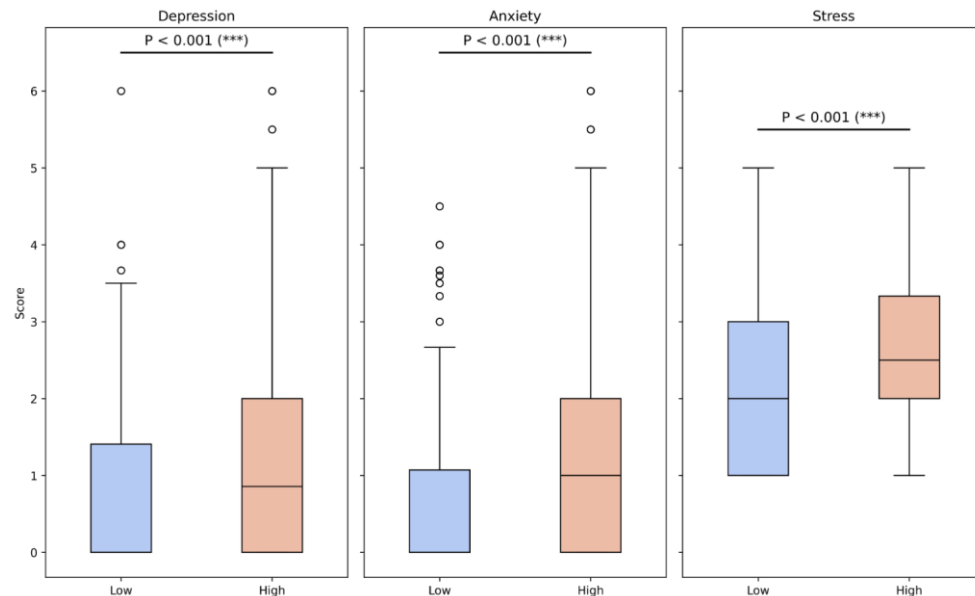
However, the relationship between emotional wellbeing risk scores (depression, anxiety, stress) and home IoT data **varied widely across individuals**

*The correlation heatmap of **users (X-axis, user ID)** and **sensor data (Y-axis)** and for emotional wellbeing risk scores

Findings for RQ1. Exploring Home IoT Data Associated with Emotional Wellbeing

Despite individual differences, **behavioral variability** emerged as a key indicator of emotional wellbeing

Greater day-to-day variability in routines was consistently associated with higher depression, anxiety, and stress levels, indicating **poorer mental health**

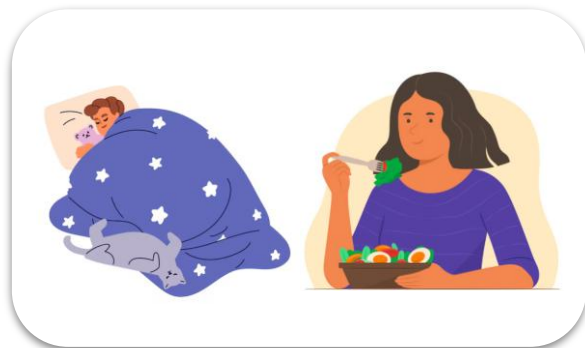


Note: *** $p < .001$, ** $p < .01$, * $p < .05$.

Findings for RQ2. User Perceptions of Home IoT Data for Emotional Wellbeing

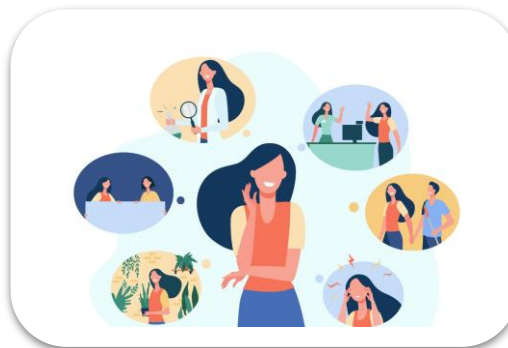
Participants replied that **the process of understanding their data** with the system was **straightforward** and **clear**

Explored **familiar or intuitive data** first,
such as sleep and eating



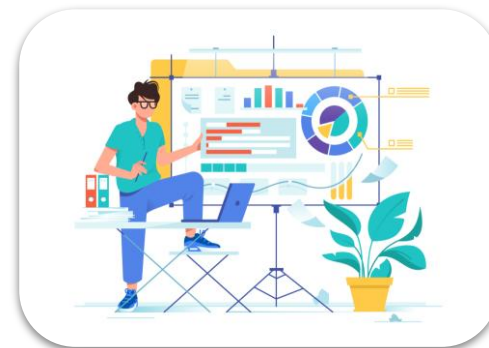
"I already know that sleep affects my mood and mental health, so I took a look at it first"

Contextualized data patterns
with **personal experience**



"...Looking back, I stayed up all night studying during exams, which added more stress..."

Integrated multiple data to
understand emotional wellbeing



"...Seeing vacuum, motion, and washer usage data together revealed how my cleaning routines were tied to my stress..."

Findings for RQ2. User Perceptions of Home IoT Data for Emotional Wellbeing

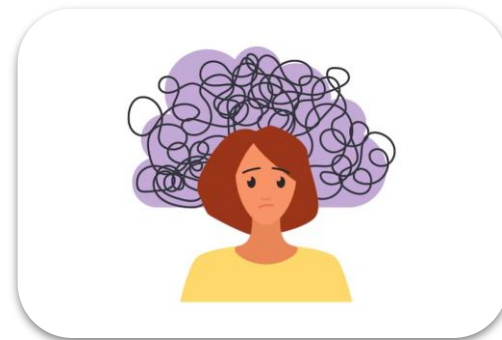
However, participants sometimes **struggled to link sensor data with emotions**

Ambiguous activities and **confounding factors**
made interpretation difficult



*"...But I often open the microwave just to clean it,
and it's still counted as usage,
even though it wasn't actually used for cooking"*

Some lacked a **clear mental model** for linking
environmental data to emotions



*"Knowing that higher humidity is associated
with my stress is interesting, but I'm not sure
what I'm supposed to do with that information..."*

Findings for RQ3. Privacy Concerns of Home IoT Data for Self-Tracking Emotional Wellbeing

After viewing their data, participants became more comfortable with Home IoT data collection. Especially, **Sleep** and **Indoor Movement** showed significant increases in acceptability ($p < .01$)

Higher perceived benefits
lead to lower privacy



*"At first, I questioned **why sleep data was needed**, but later I saw **how it reflected my mental health**"*

Understanding how data are collected
reduces privacy concerns



*"I was worried about motion data at first, but **seeing it only showed movement, not location**, made me comfortable"*

Design Implications

Based on these findings, we highlight three **design implications**:

Personalized Wellbeing Modeling



Tailor emotional insights to **individual lifestyles and routines**

Context-Aware Reflection Support



Help users **interpret abstract data** through **relatable context**

Trust-Building Privacy Design



Foster **transparency** and **control** to reduce privacy concerns

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Takeaways

- Home IoT data revealed strong **individual differences** in emotional patterns
- **Behavioral variability** emerged as a key indicator of emotional wellbeing risk
- Participants became more **open to data collection** by understanding its value and collection



Paper QR code